

Alexandra Maria Proca

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EDUCATION

Imperial College London <i>Doctor of Philosophy in Theoretical Neuroscience and Machine Learning; President's Scholar</i>	Oct. 2023 – Present London, UK
University College London <i>Master of Science in Machine Learning with Distinction; Dean's List Laureate</i>	Sept. 2020 – Dec. 2021 London, UK
University of North Carolina at Chapel Hill <i>Bachelor of Science in Computer Science, in Neuroscience with Honors, Minor in Music</i>	Aug. 2016 – May 2020 Chapel Hill, NC

RESEARCH EXPERIENCE

Visiting Doctoral Researcher <i>ENS Group for Neural Theory (Lab: Srdjan Ostojic)</i> Conducted research studying learning dynamics in low-rank RNNs	Oct. 2025 – June 2026 Paris, FR
Research Assistant <i>ETH Zürich Department of Computer Science (Lab: João Sacramento)</i> Conducted research studying the use of hypernetworks for meta-learning	Jan. 2022 – Jan. 2023 Zürich, CH
Master's Student <i>UCL Department of Computer Science (Lab: Jun Wang)</i> Conducted research studying the partial information decomposition of multitask neural networks across varying task settings in supervised and reinforcement learning models	May 2021 – Dec. 2021 London, UK
Research Intern <i>MIT Brain and Cognitive Sciences (Lab: Tomaso Poggio)</i> - Completed the MIT Summer Research Program for two summers - Conducted research adversarially reprogramming recurrent neural networks across task domains	June 2019 – Aug. 2020 Cambridge, MA
Research Assistant <i>UNC Department of Psychology and Neuroscience (Lab: Sylvia Fitting)</i> Conducted behavioral research in mice, studying HIV-1 Tat's effects on operant conditioning tasks and how endocannabinoids can be used to protect against assessed behavioral deficits	Jan. 2017 – May 2020 Chapel Hill, NC
Research Intern <i>UNC Department of Mathematics (Lab: Peter Mucha)</i> Conducted statistical research analyzing changes in neural morphology of infant fMRI data	May 2018 – Aug. 2018 Chapel Hill, NC
Research Intern <i>OSU Department of Computer Science (Lab: Radu Teodorescu)</i> Conducted research using Arduino accelerometers to create motion-detection gloves, designed to teach middle and high-school students basic programming skills	May 2015 – Aug. 2015 Columbus, OH

HONORS AND AWARDS

Imperial College London President's PhD Scholarship (Full PhD Tuition & Stipend)	January 2023
UCL Dean's List Laureate: Awarded to Top 5% of Graduating Class	March 2022
UCL Friends and Alumni Association Scholarship (\$20,000)	May 2020
Honors Carolina Laureate	May 2020
Graduated with Honors in Neuroscience from UNC	May 2020
David Bray Peele Memorial Research Award (\$220)	Oct. 2019
Carolina Research Scholar	Jan 2019
Lindquist Undergraduate Research Award (\$350)	Nov. 2018
Office of Undergraduate Research Travel Award (\$300)	Nov. 2018
Psi Chi Psychology Honor Society	March 2018
Honor's Carolina Membership	Sept. 2017
Sigma Alpha Lambda Honor Society	May 2017
The National Society of Collegiate Scholars	May 2017
UNC Dean's List	Dec. 2016 – May 2017

1. *Temporal superposition and feature geometry of RNNs under memory demands*. P. Sharma^{*}, **A.M. Proca**^{*}, L. Prieto, P.A.M. Mediano. **ICLR Oral**, April 2026.
2. *Learning dynamics in linear recurrent neural networks*. **A.M. Proca**, C.C.J. Dominé, M. Shanahan, P.A.M. Mediano. **ICML Oral**, July 2025.
3. *From Lazy to Rich: Exact Learning Dynamics in Deep Linear Networks*. C.C.J. Dominé^{*}, N. Anguita^{*}, **A.M. Proca**, L. Braun, D. Kunin, P.A.M. Mediano^{**}, A.M. Saxe^{**}. **ICLR**, May 2025.
4. *Flexible task abstractions emerge in linear networks with fast and bounded units*. K. Sandbrink^{*}, J.P. Bauer^{*}, **A.M. Proca**^{*}, A.M. Saxe, C. Summerfield, A. Hummos^{*}. **NeurIPS Spotlight**, December 2024.
5. *Training the next generation of NeuroAI researchers: Trainees' perspectives*. A. Luppi^{*}, J. Achterberg^{*}, S. Schmidgall, I. Poyraz Bilgin, P. Herholz, M. Sprang, B. Fockter, A. Siyoon Ham, S. Thorat, R. Ziaei, F. Milisav, **A.M. Proca**, H. M. Tolle, L. Suárez, P. Scotti, H. M. Gellersen. **Nature Communications**, October 2024.
6. *Synergistic information supports modality integration and flexible learning in neural networks solving multiple tasks*. **A.M. Proca**, F.E. Rosas, A.I. Luppi, D. Bor, M. Crosby^{*}, P.A.M. Mediano^{*}. **PLoS Computational Biology**, May 2024.
7. *Discovering modular solutions that generalize compositionally*. S. Schug^{*}, S. Kobayashi^{*}, Y. Akram, M. Wołczyk, **A.M. Proca**, J. Von Oswald, A. Steger, R. Pascanu, J. Sacramento. **ICLR**, May 2024.
8. *Jack of All Trades, or Master of One: Information Decomposition Reveals Distinct Features of Generalizable vs. Specialized Neural Representations*. **A.M. Proca**. Masters Thesis, University College London, London, UK, December 2021. (Supervisors: M. Crosby, P.A.M. Mediano; Advisor: J. Wang)
9. *Establishing a Contextual Fear Conditioning Paradigm for the Tat Transgenic Mouse Model*. **A.M. Proca**. Bachelors Honors Thesis, University of North Carolina at Chapel Hill, NC, USA, May 2020. (Supervisor: I.R. Jacobs, Advisor: S. Fitting)
10. *Inhibitory control deficits in Tat transgenic mice using the Go/No-Go task*. I.R. Jacobs, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, K. Mackie, A.H. Lichtman, B. Ignatowska-Jankowska, S. Fitting. **Journal of Neuroimmune Pharmacology**. 13, S38-S38 (2018).

^{*,**} Equal contribution

CONFERENCE AND WORKSHOP TALKS

1. *Temporal superposition and feature geometry of RNNs under memory demands*. P. Sharma^{*}, **A.M. Proca**^{*}, L. Prieto, P.A.M. Mediano. Oral presentation at **ICLR**, Rio, Brazil, April 2026.
2. *Learning dynamics in linear recurrent neural networks*. **A.M. Proca**, C.C.J. Dominé, M. Shanahan, P.A.M. Mediano. Oral presentation at **ICML**, Vancouver, Canada, July 2025.
3. *Learning dynamics in linear recurrent neural networks*. **A.M. Proca**. Invited talk at **COSYNE Workshop on "Causal perturbation based approaches to uncovering neural dynamics"**, Mont Tremblant, Canada, March 2025.
4. *Learning context representations in linear networks*. **A.M. Proca**^{*}, J.P. Bauer^{*}, K. Sandbrink^{*}, A. Hummos. **Junior Scientists Workshop on Recent Advances in Theoretical Neuroscience**, Trieste, Italy, June 2024.
5. *Linking generalizable intelligence to consciousness via information synergy*. **A.M. Proca**. **Association for Mathematical Consciousness Science: Modelling Consciousness Workshop**, Dorfgastein, Austria, August 2022.

6. *Fast deep learning with a simple model of the prefrontal cortex.* **A.M. Proca**, M. Wolczyk, D. Zhao, S. Kobayashi, S. Schug, J. von Oswald, J. Sacramento. **Sinergia Meeting**, Bern, Switzerland, July 2022.

CONFERENCE AND WORKSHOP POSTER PRESENTATIONS

1. *Learning dynamics in linear recurrent neural networks.* **A.M. Proca**, C.C.J. Dominé, M. Shanahan, P.A.M. Mediano. **Bernstein Conference**, Frankfurt, Germany, September 2025.
2. *Learning dynamics in linear recurrent neural networks.* **A.M. Proca**, M. Shanahan, P. Mediano. **Conference on Cognitive Computational Neuroscience (CCN)**, Boston, Massachusetts, August 2024.
3. *How context representations emerge during training: a linear network perspective.* **A.M. Proca***, K. Sandbrink*, J. Bauer*, A. Hummos. **COSYNE**, Lisbon, Portugal, February 2024.
4. *Synergistic information supports modality integration and flexible learning in neural networks solving multiple tasks.* **A.M. Proca**, F.E. Rosas, A.I. Luppi, D. Bor, M. Crosby, P.A.M. Mediano. **Analytical Connectionism Summer School**, London, England, August 2023.
5. *Synergistic information supports modality integration and flexible learning in neural networks solving multiple tasks.* **A.M. Proca**, F.E. Rosas, A.I. Luppi, D. Bor, M. Crosby, P.A.M. Mediano. **CCN**, Oxford, England, August 2023.
6. *Jack of All Trades, or Master of One: Distinct Features Between Generalizable and Specialized Artificial Neural Representations.* **A.M. Proca**, M. Crosby, P. Mediano. **Association for the Scientific Study of Consciousness (ASSC)**, Amsterdam, Netherlands, July 2022.
7. *A Picture is Worth 784 Characters: Adversarially Reprogramming a Neural Network.* **A.M. Proca**, A. Banburski, T. Poggio. **MIT Summer Research Programs Poster Session**, Cambridge, MA, USA, August 2019.
8. *Time-Dependent Inhibitory Control Deficits in Female Tat transgenic mice in the Go/No-Go Task.* **A.M. Proca**, I.R. Jacobs, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, C.B. Russell, C. Manjarres, A. Lichtman, B. Ignatowska-Jankowska, S. Fitting. **Society for Neuroscience**, San Diego, CA, USA, November 2018.
9. *Cannabinoid receptor type 1 upregulation of the infralimbic cortex of female Tat transgenic mice following ten months of Tat expression and testing for inhibitory control deficits using the Go/No-Go task.* I.R. Jacobs, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, K. Mackie, A. Lichtman, B. Ignatowska-Jankowska, S. Fitting. **Society for Neuroscience**, San Diego, CA, USA, November 2018.
10. *Time-dependent effects of Tat on Go/No-Go performance.* I.R. Jacobs, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, S. Fitting. **South Eastern Association for Behavior Analysis**, Chattanooga, TN, USA, October 2018.
11. *Inhibitory control deficits in HIV-1 Tat transgenic mice are sex dependent and alter CB1R expression.* A.B. Ferguson, I.R. Jacobs, D.J. Hermes, A.G. Antonucci, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, K. Mackie, A.H. Lichtman, B.M. Ignatowska-Jankowska, S. Fitting. **South Eastern Association for Behavior Analysis**, Chattanooga, TN, USA, October 2018.
12. *HIV-1 Tat transgenic mice show inhibitory control deficits in the Go/No-Go task.* S. Fitting, I.R. Jacobs, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, C. Xu, K. Mackie, A.H. Lichtman, B. Ignatowska-Jankowska. **Federation of European Neuroscience Societies**, Berlin, Germany, July 2018.

13. *Changes of the endocannabinoid system in HIV-1 Tat transgenic mice*. I.R. Jacobs, C. Xu, D.J. Hermes, A.G. Antonucci, A.B. Ferguson, K.L. Leggette, N.R. Miseo, **A.M. Proca**, C.B. Russell, C. Manjarres, C. Xu, M.J. Niphakis, B.F. Cravatt, K. Mackie, A.H. Lichtman, B. Ignatowska-Jankowska, S. Fitting. **International Cannabinoid Research Society**, Leiden, Netherlands, June 2018.

SUPERVISION

Pratyaksh Sharma: MEng Computing thesis on *Analyzing the computational role of complex representations in RNNs*

Nicolas Anguita: MEng Joint Maths & Computing thesis on *Learning dynamics of linear neural networks*

TEACHING

Interpretable and Explainable AI <i>ENS-PSL Week</i>	Nov. 2025 Paris, FR
Computational Neurodynamics <i>Imperial College London</i>	Oct. 2024 – Dec 2024 London, UK
Introduction to Research in Network Data Science <i>UNC Chapel Hill</i>	Jan. 2019 – May 2019 Chapel Hill, USA

REVIEWING

International Conference on Learning Representations (ICLR) 2026

International Conference on Machine Learning (ICML) 2025

Conference on Cognitive Computational Neuroscience (CCN) 2023, 2024

ORGANIZATIONAL ROLES

Organizer for <i>UniReps</i> <i>ELLIS x UniReps Speaker Series, NeurIPS Workshop</i>	Jan. 2026-Present
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TRAINING AND WORKSHOPS

Junior Scientists Workshop on Recent Advances in Theoretical Neuroscience	June 2024 Trieste, IT
Gatsby Unit Analytical Connectionism Summer School	August 2023 London, UK
Mathematical Consciousness Science: Modelling Consciousness Workshop	August 2022 Dorfgastein, AUT
Sinergia Meeting 2022	July 2022 Bern, CH
UCL & PSL Summer School on Consciousness and Metacognition	June 2021 London, UK
MIT Brains, Minds, and Machines Summer Course	Aug. 2020 Cambridge, USA
University of Nicosia Summer Pre-Med Program	June 2017 – July 2017 Nicosia, CY

OUTREACH

Invited speaker for Imperial Computing Undergraduate Society (DoCSoc) <i>Lecture on mechanistic interpretability</i>	2025 London, UK
Invited speaker for Girls Who Code at Harriton High School	2024 Bryn Mawr, PA
Graduate student panelist for UNC Neuroscience majors	2021 Chapel Hill, NC

EXTRACURRICULAR AND SERVICE

Member <i>Mechanistic Interpretability Journal Club</i>	Sept. 2024 – Present <i>London, UK</i>
Organizer <i>Qualiaheads Consciousness Science Journal Club</i>	Feb. 2022 – Jan. 2023 <i>Zürich, CH</i>
Treasurer <i>Psi Chi Psychology Honor Society</i>	March 2018 – May 2019 <i>Chapel Hill, USA</i>
Executive Officer <i>Carolina Neuroscience Club</i>	Jan. 2018 – May 2020 <i>Chapel Hill, USA</i>
Piano Instructor <i>Musical Empowerment</i>	Sept. 2017 – May 2020 <i>Chapel Hill, USA</i>
Member <i>Women in Computer Science</i>	Sept. 2017 – May 2020 <i>Chapel Hill, USA</i>
Boston Qualifying Team <i>UNC Marathon Team</i>	Sept. 2016 – May 2020 <i>Chapel Hill, USA</i>
Swim Coach <i>Worthington Special Olympics</i>	Sept. 2014 – Aug. 2016 <i>Columbus, USA</i>

SKILLS

Python, Pytorch, JAX, Java, Latex, C/C++

LANGUAGES

English: Native
Romanian: Conversational
French: Elementary
Spanish: Elementary